Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2025-26 (EVEN)**	**Course Name**	**LAB – Reinforcement Learning**
Semester	6th	**Course Code**	**23ADS1644**
Roll No	**42**	**Name of Student**	**Krishna Mishra**

Practical Number	**04**
Aim	To apply K-Means clustering to group data points into clusters based on similarity with and without labeled output.
Problem Definition	To implement K-Means clustering for grouping data points into clusters based on similarity, and to analyze clustering results with and without labeled data.
Theory (100 words)	K-Means clustering is an unsupervised machine learning algorithm used to group data points into a predefined number of clusters based on similarity. The algorithm partitions the dataset into *K* clusters, where each data point belongs to the cluster with the nearest mean (centroid). It works iteratively by initializing centroids, assigning data points to the nearest cluster, and updating centroids until convergence is achieved. The objective of K-Means is to minimize the within-cluster sum of squares (WCSS), ensuring that data points in the same cluster are as similar as possible. It is widely used for pattern recognition, customer segmentation, and data compression. Although K-Means is an unsupervised technique, labeled data can be used to evaluate clustering performance by comparing predicted clusters with actual labels. Visualization techniques such as scatter plots help in understanding cluster distribution. However, K-Means assumes spherical clusters and is sensitive to the choice of K and initial centroid selection.

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Procedure and Execution (100 Words)	Algorithm: 1. Import required libraries (NumPy, Pandas, Matplotlib, sklearn). 2. Load the dataset from a local file. 3. Perform data preprocessing (handle missing values, normalize if required). 4. Select relevant features for clustering. 5. Choose the number of clusters (K) using methods like the Elbow Method. 6. Initialize and apply the K-Means algorithm to the dataset. 7. Assign each data point to the nearest cluster centroid. 8. Visualize the clusters using plots and compare with labeled data (if available). 9. Analyze clustering results.

Code:

```
#  K-Means Clustering ---


import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans,
AgglomerativeClustering
from sklearn.preprocessing import
StandardScaler
from scipy.cluster.hierarchy import
dendrogram, linkage


# Load dataset
data = load_iris()
X = data.data

# Normalize data
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)

# Elbow Method to find optimal K
wcss = []
for k in range(1, 11):
    kmeans = KMeans(n_clusters=k,
random_state=42, n_init='auto')
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

# Plot Elbow curve
plt.figure(figsize=(8, 4))
plt.plot(range(1, 11), wcss, marker='o')
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()

# Apply K-Means with optimal K (3)
kmeans = KMeans(n_clusters=3, random_state=42,
n_init='auto')
clusters = kmeans.fit_predict(X_scaled)

# Visualize K-Means clusters (First two
features)
plt.figure(figsize=(8, 4))
plt.scatter(X_scaled[:, 0], X_scaled[:, 1],
c=clusters, cmap='viridis')
plt.xlabel("Feature 1 (Sepal Length)")
plt.ylabel("Feature 2 (Sepal Width)")
plt.title("K-Means Clustering")
plt.show()
```

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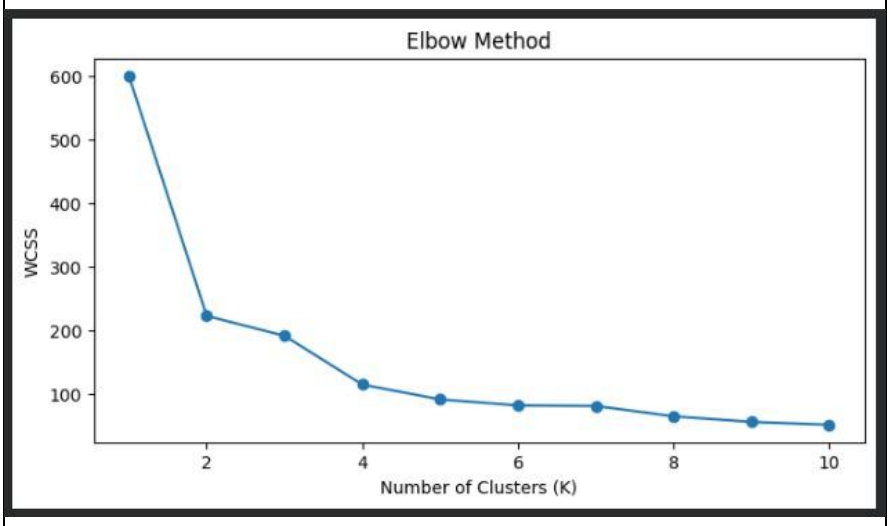
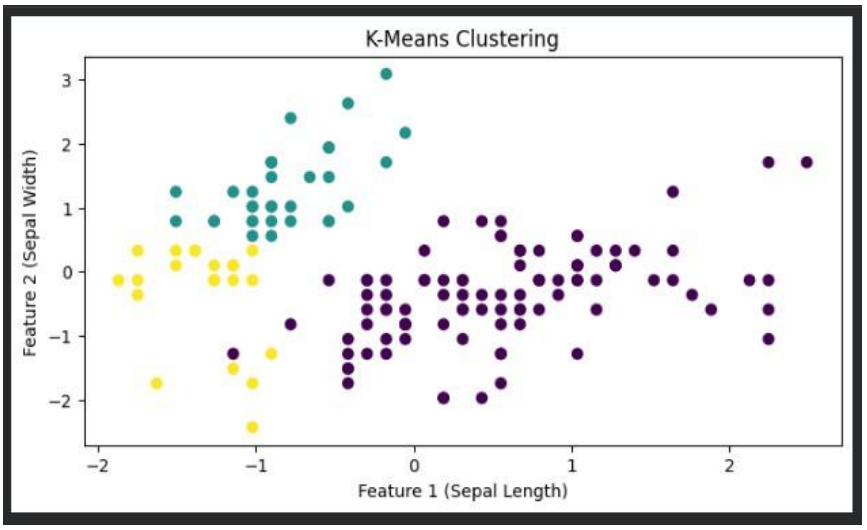
Output:





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Conclusion	K-Means clustering effectively groups similar data points into clusters, helping to identify hidden patterns in data. It is simple, efficient, and useful for exploratory data analysis, though its performance depends on the choice of K and data characteristics.
Date	**28-03-2026**